



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic:Conditional Probability.....

LECTURE -10

EXPLANATION

Conditional Probability - Let A and B are two events associated with some random experiment. Then the probability of occurrence of event A when event B has already occurred is called the conditional probability.



$P(A/B)$ = Probability of event A given that B has already occurred.

$P(B/A)$ = Probability of event B given that A has already occurred.

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

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Examples -

Q1) A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once.?

Ans - $2/5$

Q2) A coin is tossed three times. Find $P(E/F)$, if the events E and F are as follows.

(i) E: Head on third toss F: heads on first two tosses

(ii) E: Atleast two heads F: Atmost two heads

(iii) E: Atmost two tails F: Atleast one tail

Q3) A die is rolled. If the outcome is an even number, what is the probability that it is a prime number?

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